

RATIONALES OF THE SPACE PROGRAM

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INTRODUCTION

For the believer in outer space as the next human frontier, the "why space" question, the rationale for exploring space, is self-contradictory. The very existence of space as an unexplored and, until recently, an inaccessible physical location justifies the cause for exploring space. One view holds that humans, being curious and adventurous as a species, naturally reach out to explore, exploit, and utilize new and unknown frontiers. Another view is that exploration is not necessarily natural to the human experience. Space as an object of interest fascinates the public; however, the public when asked to choose between expanding the United States (US) national space exploration effort and doing other socially relevant tasks, most citizens consistently choose the latter category. Space has remained ancillary US public policy for most of its history.¹

This reality means that continued pursuit of a viable US civil space program must incorporate a number of goals and rationales. What has occurred historically is that the US, usually through NASA, has expounded a rationale for the exploration of space within which are nested several diverse and complementary facets.² Those various facets rise and fall in importance and intensity with which they are pursued across the decades as US space policy responds to dramatic changes in the international and domestic political climates. This chapter sketches out that evolutionary path, focusing upon the past and present as a prelude to the future. Individuals often disagree as to the relative weight that should be assigned each facet at a particular time. The public disagreements usually reflect differences not in desired goals but in the intensity with which each are being pursued. For example, disputes have arisen over the relative importance assigned human spaceflight in comparison to space science undertaken by robotic probes.

One must also acknowledge that there also exist those whose viewpoints reflect continual rejection of any significant governmental civil space program. Given the increasing ubiquity of space applications directly applicable to daily life, such a perspective might be more supportable today as

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the commercial space sector expands to meet the burgeoning demand for their services.³ Earlier, such a perspective made less sense, as no independent commercial space sector existed. Those in this sector reject human spaceflight as just another costly foreign policy driven venue that is wasteful of resources.⁴ Others saw space programs as nothing more than an issue of prestige, a monument to the modern state and human pride. Albeit these views are counter justifications for a civil space program, they have influenced the process by which the rationales for such a program have been constructed and have played a role in the evolution of US space policy.⁵

FRAMING THE ANALYSIS

In considering rationales, one must begin with the clear understanding that the entire US civil space enterprise originated in the public sector. Whenever one desires to expend public funds for a public purpose, the reasons advanced must be persuasive across a broad spectrum of political factions and advocacy coalitions, including both potential supporters and opponents. The simple fact is that space expenditures are minimal compared to defense or social expenditures. This difference can be seen annually in the arguments advanced in Congress concerning NASA expenditures. A believer in space's enormous scientific and economic potential may feel that space by its very existence justifies such tremendous expenditures. Concomitantly, public officials operate in a larger political context within which other equally worthy causes and their group representatives, such as advocacy coalitions, stridently demand their full attention. This political competition is exacerbated by the fact that governmental resources for a particular public policy are usually limited and must be spread across multiple actors and policies.

The existence of such competition means that the public rationales advanced must clearly be persuasive at several levels of argument. Rationales must meet the perceived needs of relatively large and diverse economic and political constituencies. Advocates have to create needs or meet expectations about the future. This is especially critical for a new field as space exploration initially was. Immediate policy successes do occur but long-term financial commitments ultimately determine the degree of success achieved. Consequently, no single rationale or justification can meet the political needs; rather, different facets are emphasized at various points in time.

Given both the large public sums required and the initial vagueness of any promised results, a particularly strong political incentive was required to push space efforts forward. The vagueness of results implies that until space was actually physically occupied on a continuing basis, it remained somewhat

unclear what exactly could be accomplished. From 1945 onward, Earth-orbiting satellites were considered realistic; their successful employment required real time experience rather than hypothetical conjecture.⁶ Space's physical environment raises unanticipated hazards, continually demonstrating the inadequacies of earlier solutions. For example, satellites encounter solar flares and orbital debris outside the parameters of what is expected. Space activity remains a continual learning experience.

Space advocates pushed forward a variety of purposes for exploring and utilizing space, part commercial, part scientific, and part military, although all these activities were wrapped up in the overriding vision of humans exploring space. Even though human spaceflight was central to the vision, it clearly remains the most politically challenging and expensive task to achieve successfully. Further, once humans along with their machines arrived in space, the future remained less than clear regarding the actual benefit to society. There was the honor and thrill of being "first" to land a man on the Moon, but why one should go back was the real question. Being first was an expensive political prize of the Cold War if that was all that occurred.

One needs an incentive, a compelling focusing event, strong enough to break through the existing political status quo and to place the issue of space on the policy agenda for political decision-making and policy formulation. The lack of a focusing event delayed the US from orbiting a satellite first. This was so despite the fact that programmatic options and the technology for space activity existed in the early 1950s. Clearly, other political priorities intervened which prohibited taking effective action prior to the Soviet Union's launch of the world's first orbiting satellite in 1957.

The focusing event came with the October 1957 Soviet launch of Sputnik 1 followed by an even greater mass Sputnik 2. Sputnik was part of the Soviet participation in the International Geophysical Year (IGY). The US had also agreed to participate in IGY through its quasi-civilian Vanguard "satellite" program. IGY was a scientific program organized on the support of national governments throughout the world. Its scientific goal was to survey the geophysical environment of Earth from outer space. From the beginnings of the space age an important rationale for US space activity became scientific exploration. This began with exploration of Earth's near space environment and expanded beyond Earth orbit to encompass other celestial bodies and astronomical observations of the cosmos.

The US Earth-orbiting satellite program carried a dual burden: scientific exploration of outer space and critical national security implications. From its inception the US space program has been immersed in international politics and served several purposes simultaneously, including the legal

precedent involved in establishing the right of free passage for satellites in Earth orbit. Airplanes violating national air space are subject to interception and possible destruction. Satellites in the 1950s represented a particular unknown as to the legal issue of the extension of sovereignty over air space to outer space since at that time no country possessed the technical capability to intercept an orbiting satellite. The possibility of diplomatic protest denying the political acceptance of unimpeded overflight by satellite existed. This implied that later when an antisatellite capability was achieved, interception would follow as a matter of course. The Sputnik successes were followed in January 1958 by the US Explorer I satellite; both occurred without public or governmental protest and answered the legal question of overflight. Peaceful free passage in space for all nations became the legal norm and custom, and by 1967 was codified as one of the legal foundations of the Outer Space Treaty.

Free passage implies that all satellites pass above the Earth's surface without hindrance. As such, international regulation of satellites was necessary. An elaborate set of policies and rules gradually came into force under the auspices of the International Telecommunications Union (ITU) to prevent interference by different satellites regarding transmissions from other satellites and their passage through space. Later, more elaborate rules were established via the International Telecommunications Satellite Organization (Intelsat) allocating scarce economic assets such as satellite location along the orbital arc (orbital slots) and geosynchronous orbits. Operating within these international understandings became the legal norm.

In terms of justifying a US space program, the two Sputniks provided a sufficiently strong political crisis to create intensified support for that effort. The Soviet government pushed hard on the theme that Sputnik established without equivocation their ability to threaten American security.⁷ As a consequence, American space policy was buffeted by the squalls of domestic public opinion, demanding something be done. This episode created the strong pressures for making space policy an adjunct of national security policy rather than emphasizing the pursuit of peaceful activities, such as scientific exploration and commercial development. The concern for national security drove the field for years, and still remains an important issue. What occurs in American space policy today, relative to the rationales supporting the overall program, grew out of that formative experience— one shaping both expectations and institutions. In fact, NASA itself arose out of that Cold War crucible.

The rationales articulated have appeared in a variety of forms: policy statements and speeches by public officials including the President; policies by NASA and other government players; laws passed by Congress; policy statements by national commissions; and visions of the future by space

advocacy coalitions. The space program is surrounded by political groups interested in influencing its future directions. Industry and the various scientific societies and research organizations have a material and an intellectual interest in what occurs. Advocacy coalitions come to the table with diffuse and at times competing motivations. Government participants have historically been influential because they provide the resources, leading the field in particular directions.

NASA's outreach activities are responsible for getting the Agency's message out to the public.⁸ The goal is to articulate the message in an interesting and relevant form to the public who ultimately fund the Agency's work through tax dollars. These activities can involve both curricular support of K-12 programs and broader community outreach forums through local groups and schools. Regardless, there are countervailing players, especially the commercial sector, whose views differ from those espoused by NASA. The rationales supporting the US space program have arisen from the interplay between these diverse and often competing interests. At times, NASA feels that its particular message gets lost in the cacophony of politics.

Table 2.1 provides examples of major public statements regarding space activities. Some are critical, as is the Rogers Commission report regarding the Challenger accident directed at putting the American Space Shuttle program back on track. The first two examples, *Collier's* and the Disney television feature on "Man in Space," represent unofficial expressions impacting public policy debate. Indeed, both *Collier's* and the Disney feature created an atmosphere conducive to supporting a larger vision of human space activity.

President Kennedy's speech before Congress in May of 1961, where he advocated landing a man on the Moon by the end of the decade of the 1960s, serves as an expression of presidential leadership that drove US space policy for a decade. As can be seen from the listing in Table 2.1, the more usual statements are presidential policy directives and reports by advisory groups of differing pedigrees. Most Presidents since Eisenhower sign formal statements of their administration's space policy. Advisory committees are a time-honored method of presidential advice seeking. In some instances the committee recommendations are superseded by a change in administration (e.g., Ad hoc Panel on Man-In-Space) or by events (e.g., Pioneering the Space Frontier). Others just have their recommendations rejected by the administration because the report does not meet its needs (e.g., Space Task Force).

Table 2.1. Public Policy Statements Regarding Space Activities.

Author	Title	Date
Editors, <i>Collier's</i>	"What Are We Waiting For?"	October 1952
Walt Disney	"Man In Space"	March 1955
National Security Council (5814)	"US Policy On Outer Space"	August 1958
President's Science Advisory Committee	"Report of the Ad-hoc Panel on Man-in-Space"	December 1960
President Kennedy	"Urgent National Needs" (Speech to Congress)	May 1961
NASA	Summary Report: "Future Programs Task Group"	January 1965
Space Task Force	"The Post-Apollo Space Program: Directions for the Future"	September 1969
President Carter	Presidential Directive: "Civil And Further National Space Policy"	October 1978
President Reagan	"National Security Decision Directive: National Space Policy"	July 1982
NASA	"NASA Commercial Space Policy"	October 1984
Presidential Rogers Commission	"Report of the Presidential Commission on the Space Shuttle Challenger Accident"	June 1986
National Commission	"Pioneering the Space Frontier"	May 1986
NASA Task Force (Ride Report)	"Leadership and America's Future in Space"	August 1987
President Reagan	Presidential Directive: "National Space Policy"	February 1988
President Bush	"Space Exploration Initiative"	July 1989
President Bush	"Commercial Space Launch Policy"	September 1990
Augustine Committee	"Report of the Advisory Committee on the Future of the US Space Program"	December 1990
President Clinton	"National Space Transportation Policy"	September 1994
President Clinton	"National Space Policy"	September 1996

Any suggestions supporting space activity and setting future directions run a gauntlet of political and fiscal practicality problems that are an inherent part of public policy. Over time, political justifications decline in relevance or events remove them from the public policy agenda altogether.

Thus, each presidential administration, even if disinterested (e.g., the Carter Administration), must develop a rationale for space that fits its policy agenda and political needs. This requires a continuing process of policy adjustment and change. Space advocates inside and outside the government are often frustrated by ongoing policy change that has made and continues to make the formulation and implementation of space programs problematic.

STABILITY AND CHANGE

Current space activities are outgrowths of rationales, programs, and trends set in motion years beforehand. Presidential administrations build on the handiwork of the past for several reasons: the issues have been thoroughly researched, thought about, and hashed out by bright people; the obvious has been discovered, considered, used, or discarded (the latter usually for good reason); space activity is a large-scale endeavor; and previous decisions structure future choices because budgets are limited. Revamping programs is unlikely to be a productive choice as one moves the US political system incrementally in new directions.

Rationales for US space policy have fallen into distinct clusters as depicted in Table 2.2. The argument is that early events laid the basis from whence flowed both the specific programs and activities operating today along with their supporting rationales. This is a capsule view of the space age strictly from an American perspective.

Table 2.2. Rationales and Space Policy.

Time Line: Activity:	Beginnings	Present	Future
Military	Military Space	Force Support Force Enhancement	Space Control Force Application
Scientific	Space Science	Earth Science Astronomy	Planets Asteroids Environment
Civil	Human Spaceflight	Space Shuttle Space Station	Human Habitation Human Exploration
Commercial	Technology Development	Commercial Applications	Economic Competitiveness

In Table 2.2, the timeline moves forward from the beginnings to the present and into the future helping to place different events into their proper historical sequence. The dawn of the space age begins in 1957 as Sputnik entered orbit, represented by "Beginnings," a period running roughly across the first years of the space age, the Apollo-era. The "Present" encompasses the post-Apollo period, while the "Future" is located in the twenty-first century. Unlike Sputnik's launch, subsequent time periods, the present and future, are not clearly delineated by a singular focusing event, instead they reflect trends that emerge into prominence or fade away on policy agendas. The point here is to provide a broad perspective for the historical movement of events and the supporting rationales justifying the specific activities.

American space activity whether military, scientific, civil, or commercial emerged originally from within the driving issue of national security. This issue has been designated the "prime mover," meaning that public concern regarding this question was sufficiently strong to generate intense political demands for an expansive American space effort. Despite great reluctance on the part of President Eisenhower, he was politically pressured to expand American civil space efforts to include human spaceflight.⁹ National security, operating as the justification for space activity, reflected the historical context of the Cold War, an international political-military-economic competition occurring between blocs led by the US and the Soviet Union respectively. In 1957, those alliances were locked in an intense duel for world leadership; space activities were incorporated into that competition because of their military relevance.

The rocket that placed a satellite in orbit also possessed the capacity to deliver a nuclear warhead. Accordingly, falling behind in peaceful space activities was perceived as evidence of lagging behind in the competition for strategic nuclear missile superiority. The fallacy of that view became only clear in retrospect.¹⁰ Sputnik and the political fallout from that event created incredible political pressures to be "first" and technologically ahead of the Soviets. Only after those events, with the ending of the Cold War and collapse of the Soviet Union, did the original national security rationale for a national civil space program become secondary, allowing for fuller articulations of other rationales. National security has never faded out of the picture, but the emphasis has become less military and more concerned with economic competitiveness.

Since Sputnik, four major themes arose to justify a large-scale American space policy. Those themes have played out in the past and will continue in different forms into the future. Specific programs have fluctuated over time but were worked out within the boundaries of these general interests. In Table 2.2, the intention is to provide some examples of programs whose justification flows from the larger theme in which they are embedded.

Although this chapter focuses primarily on scientific, civil, and commercial space activities, military space underlines all of the other activities since civilian space and commercial space often flowed out of military endeavors.

In addition to the four major rationales underlying US space efforts, explicit mention must be made of the sub-theme of international space cooperation. Internationalization of space activities was enunciated as a goal from the beginnings of the space age (e.g., IGY). The National Aeronautics and Space Act of 1958 establishing NASA contained the statement: "...cooperation by the US with other nations and groups of nations in work done pursuant to this Act."¹¹ Interactions with potential international partners, such as the Soviet Union, were limited by Cold War politics. Adversaries were perceived as unlikely partners, notwithstanding rhetorical gestures that were made soliciting their cooperation. President Kennedy, even after proclaiming the Apollo Program as a national priority, sought later before the United Nations (UN) to jointly fly to the Moon with the Soviets.¹² Concurrently, given the strategic military balances of the Cold War, cooperative offers were made toward US allies. Those efforts began in the 1960s, but were always restricted by national security and technology transfer concerns.¹³ Nevertheless, from the beginnings of the space age international cooperation was an integral part of space activities, especially regarding space science programs.¹⁴ The current multilateral cooperation on the International Space Station (ISS) became an extension and amplification of space cooperation.

Military Space Activity

Military space activity has proceeded through several stages in development. From the beginning, securing the US from nuclear holocaust was the primary goal. Initially, the issues involved establishing a sufficient reliable military space presence to enhance the effectiveness of American strategic nuclear forces.¹⁵ That effort led to other efforts aimed at developing elaborate missile detection and warning systems along with a reconnaissance capacity sufficiently accurate to override Soviet secrecy. Finding targets behind the Communist "Iron Curtain" was an absolute priority if American nuclear deterrence was to work. The space station depicted in the Disney feature listed in Table 2.1 was a reconnaissance system by which the Soviet Union and the rest of the globe could be observed.

From the Eisenhower Administration onward, military space activities have largely been conducted in secrecy or at least within a low profile, minimizing public visibility and awareness of the extent of those

activities.¹⁶ One important thrust has been the extension of military command and control functions to the theater level. The earlier space-borne assets were reconfigured to provide force support and force enhancement at all levels of the military activity spectrum. More recently, the US military has begun examining and testing space-based technologies aimed at protecting US space assets, whether they are military or nonmilitary, while creating the capacity to deny an enemy access to orbit.¹⁷ This involves space control and force application.

Scientific Space Activity

Achieving access to space attracted serious attention from the beginning because of its potential for expanding human knowledge about the Earth and the cosmos. Satellites in Earth orbit provide insight into the physical processes affecting the Earth. Several themes have been built into this rationale extended to support space science activity. At one level, there exists the idea of the pursuit of abstract knowledge—knowing more about the Earth and cosmos in order to expand the human mind.¹⁸ This can be seen in the first reason advanced in the NASA statute: "...the expansion of human knowledge of phenomena in the atmosphere and space."¹⁹

Space science has continually drawn verbal and fiscal congressional and presidential support but it has proven less important than the pursuit of knowledge capable of some social or economic payoff. Consequently, since the earliest days the rationales have equally emphasized the likely rewards for pursuing such knowledge: "...establishment of long-range studies of the potential benefits to be gained from...the utilization of aeronautical and space activities for peaceful and scientific purposes."²⁰ Originally, the benefits to be obtained were somewhat unclear because no one really knew whether long term operations in space were possible.

More recent expressions regarding this theme have become explicit as the emphasis shifted to Earth science, especially in regard to the environment. NASA's Earth Science Enterprise, justified by the need to establish the social relevance of space activities, assesses global change phenomena. If governments are to make informed decisions concerning critical environmental issues, a better knowledge base must be demanded to support those decisions. The acquisition of knowledge can engender policy learning that could counter the strong ideological strands running through environmental policy debates over such issues as global warming. This, in turn, can lead to policy action aimed at mitigating further global environmental change.

The pursuit of knowledge in different forms has been central to the rationales advanced for US space policy. For example, space science budgets have been tapped several times to support other programs. Albeit science activities were not particularly prominent in the Apollo program, they grew in importance once the first landing on the Moon occurred.²¹ Moreover, space opened up an enormous number of avenues for further scientific research. Given cost constraints on space science programs, only a comparatively few missions and scientific questions can be pursued in any depth at a single time. Currently, robotic Mars exploration and questions of the "origins of life" are dominating the policy agenda in the space science arena.

Civil Space Activity

The centerpiece of American space activities has been the sending of humans into orbit. In early 1958, before Congress established NASA, a committee reported to the future Agency regarding the composition of the national civil space program.²² In that report, manned spaceflight was listed as a priority, a position ratified by the Agency in its first "Long Range Plan."²³ Human spaceflight has remained a major thread in civil space activities. The dream was first reaching orbit and then establishing a permanent human residence in space. The Moon came early due to a political aberration (e.g., Cold War competition), not duplicated since then.²⁴

ISS is currently the linchpin around which US crewed space efforts revolve.²⁵ Cost was clearly the stumbling block in the post-Apollo era and remains a concern. Placing humans in orbit on a continuing basis changes the dynamics of space policy creating opportunities otherwise not available, such as industrial level microgravity manufacturing. Other space-based research activities become doable because long-term exposure to the space environment, whether internal or external to the spacecraft facility, is possible. Another argument made is that the presence of humans in space opens unanticipated opportunities because human ingenuity and observational talents will create new vistas. "Man has the capability of correlating unlike events and unexpected observations, a capacity for overall evaluation of situations, and the background knowledge and experience to apply judgment that cannot be provided by instruments..."²⁶ Human flexibility was best observed in the Space Shuttle missions to service the Hubble Space Telescope where unanticipated repairs were identified and solved on the spot by the astronauts.

The dream of humans living in space has been the core theme of much of the space literature.²⁷ Critics, among them space scientists, argue

that humans clutter up the situation, making it too expensive to complete tasks that a robot could more cheaply accomplish. Politically speaking, humans in space has been an essential part of the justification for space policy in that those individuals personalize what otherwise is a distant and alien environment. From the Mercury astronauts onward, space voyagers have been a glamorous group.²⁸ Loss of human spaceflight, it is argued, crushes the dream and is a defeat for the human spirit. NASA's "Return to Flight" after the Challenger accident was in part based upon that premise; humans must move past the tragedy to reassert their destiny.²⁹ The US has incorporated such activities into the essence of its space program, sustaining its crewed flight operations despite the tremendous costs involved, operational difficulties, tragedies such as the Apollo 204 fire and the Challenger explosion, and budgetary and international political complications as has been the case with ISS.³⁰

Commercial Space Activity

Space activities are technologically challenging exercises. The successful overcoming of those challenges was in-and-of-itself deemed an important justification for the commercialization of space. Nations capable of sustaining space commercialization were major world economic competitors and are viewed in international politics as modern developed states. Space technologies demanded skilled and well-trained work forces whose talents could be disseminated into the larger technological and economic base of the nation. Apollo was developed into an economic engine for the Southern States of the US and represented a "Space Age America." This "Space Age America" idea served as powerful rationale for Apollo in that those who held this view, among them President Johnson, believed that the technology mobilized and harnessed by the government for sending men to the Moon could be applied to solve the social ills of the country. In the post-Apollo era, NASA established a series of Commercial Space Centers between industry and universities to foster space commercialization. The economic rationale has become stronger and even more explicit as space applications, such as telecommunications and remote sensing satellites, are important for sustaining US global economic competitiveness.

The Reagan Administration was the first to expound the economic impact of space applications by enlarging the private sector's role while reducing the government's role. This altered the dynamics under which commercial space activities occurred. One example of this is the "National Space Policy" promulgated in 1982. "The US encourages domestic commercial exploration of space capabilities, technology, and systems for

national economic benefit."³¹ As the US government copes with the end of the Cold War and the global war on terrorism, the issue of national security restrictions on the process of space commercialization has come to fore for the space commercial sector.³² Space technologies are becoming the means by which other high-tech applications can be facilitated. Government space activity, through the Department of Commerce and NASA, has helped the development of space commercialization. President Clinton's 1996 Space Policy and the Commercial Space Act of 1998 are important examples of public policy that facilitate this trend.

The Commercial Space Act is a landmark public policy mandating commercial activities in space that have been previously dominated and controlled by the government. These areas include Global Positioning System (GPS), acquisition of space science and Earth science data, and space transportation services. The Act calls for the government to make use of commercial space technology rather than to develop these technologies or capabilities on its own, and for the government to make use of its own space assets to promote space commercialization. NASA has in place an ISS commercial development plan, and the US military recently saved the Iridium satellite constellation from a planned de-orbit by making use of the global satellite network for its own communication needs.

CONCLUSIONS: RATIONALES AND EVOLUTION OF SPACE POLICY

The research literature that examines the rationales underpinning the US space program is fairly sparse and highly selective. This state of the literature is a function of the space policy field's development. Many of the writers engaged in space policy analysis have been policy participants and advocates, meaning their analyses are focused upon explicating and justifying their viewpoints regarding the issues of the day.³³ The most prominent writers have been direct participants in the space program including Wernher von Braun³⁴ and Carl Sagan.³⁵ Advocacy literature is typical because the policy choices are controversial. For instance, the current debate over the feasibility of reusable launch vehicles (RLVs) generates controversy between economics and technical feasibility. This occurs in advance of any demonstration of the RLV concept in actual flight, resulting in literature that focuses upon technological specifics rather than the broader political contours embodied in rationales.

Academic observers, who write in the area of space policy, put forward analyses that are either technical-analytic focused or policy-studies oriented. In the former case, the focus is on effectively accomplishing the

specific mission or overcoming a particular technical problem where political issues are external to the frame of reference. There is a disjunction between practitioners and policy-makers due to this myopia. Public policies by definition involve politics, and analyses that fail to incorporate such insights remain incomplete. In the latter case, researchers tend to concentrate on how a particular program meets its programmatic goals.³⁶ These program goals flow from the broader rationales discussed herein yet the very rationales underlying the goals are either unexamined or are assumed to be fixed and agreed-upon. Such a disconnection is typical of academic disciplines where certain fundamentals are assumed to be true; therefore, the research focuses on the factual issues generated by those rationales.

Another factor impacting the development of the space policy field is the original public policy setting in which the field developed. During the Eisenhower Administration, there was a potent counter-critique regarding what the US space program should entail in terms of rationales and programs. Kennedy's endorsement of the Apollo Program and the acceptance of a comprehensive US space program stymied that debate. Alternative views were perceived as statements by political outsiders, from both the left and the right, and the self-interested. Those who felt the monies should be used for other activities, such as social welfare, were unsuccessful. Others were dismissed as self-interested because they, usually space scientists, advocated not fundamental change but a recasting of priorities to favor scientific interests rather than human spaceflight.

The literature that tends to deal somewhat with rationales is found in histories being written concerning the space age and the US space program.³⁷ McDougall's *...the Heavens and the Earth: A Political History of the Space Age* recast how the space policy field's development is perceived.³⁸ This publication was explicitly political in its focus and comprehensive in its historical breadth. A critical evaluation of the policy choices made by President Kennedy were placed by McDougall in a broader historical context in which their impacts could be assessed relative to the US space program's long range goals and development.

Additional literature that bears on rationales is the NASA-funded series detailing the documentary history of US space exploration and development of space. Under the editorship of John M. Logsdon, a series of historical volumes and survey essays present analyses of different facets and phases of the US space program.³⁹ The importance of these volumes is the identification of the general themes supporting the extension and continuation of the American space program. Each survey essay in these volumes is followed by excerpts from official documents created at the time. The essays reflect the assessment of the authors, but those interested are provided access to the source materials that can lead to further examination. Much of the

source material regarding the creation and agenda-setting for the American space program can be found within these volumes.

Space policy literature that uses Presidents and their administrations as the unit of analysis is also useful to assess the issue of rationales. This type of analysis is based on the fact that each President approaches the space program from a different perspective as indicated in Table 2.1; Presidents and their advisors respond to ongoing political events, opportunities perceived to exist, and their evaluation of the intrinsic worth of the program. Presidents Nixon and Carter expressed great skepticism as to the value of human spaceflight, while Presidents Reagan and Bush (89) articulated grander visions, including the space station in 1984 and the aborted Space Exploration Initiative (SEI) of 1989 that called for the establishment of a lunar base and human missions to Mars.

One study by Eugene M. Emme provides an earlier perspective upon the evolution of space policy through the different presidential administrations.⁴⁰ This study, however, occurred prior to the Space Shuttle and its subsequent problems. Logsdon's survey essay in the NASA history of the US space program provides a perspective on Presidents and space policy that includes the Bush Administration (89).⁴¹ This perspective covers several crises including the Space Shuttle Challenger accident after which NASA and the space program had to reestablish credibility. An additional publication has explored the development of space policy from the perspective of whether Presidents in fact do lead the space program as it deals with the establishment of programmatic goals and priorities.⁴² This volume offers a different perspective in that analyses are written by scholars whose research specialty is on a particular President and administration. The volume provides insight into the justifications for particular space policy choices made by the different Presidents.

In all these aforementioned analyses, the rationales are only indirectly examined. As this chapter suggests, the central themes underlying the rationales of the US space program were largely established in the late 1950s. Since that time, different variations of the same basic underlying rationales in response to changing political environments have been utilized for support of space policy decisions; but the underlying rationales, such as support of a human presence in space, has not changed despite changing historical conditions and differences in presidential ideology and policy.

One book that comes closest to directly examining the rationales underlying the American space program is Howard McCurdy's volume exploring the impact of space as a concept rooted within the American imagination.⁴³ This work traces the development of the concept of space and shows how that development is linked to pulp fiction. McCurdy's analysis

illustrates the fascination technology holds on the American psychic and how space plays into that perception. Embedded in this work are the four themes articulated in Table 2.2.

The US established a hierarchy of rationales justifying its space activities. This intellectual structure has endured in an interesting way; much like a diamond, different facets of the gem glitter with greater luster at various points in history—rising and falling on policy agendas—but they remain a coherent set of ideas and concepts supporting an expanded view of the human presence in space whether actual humans or their machines.

ADVOCACY COALITIONS AND SPACE POLICY

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INTRODUCTION

Even though organized advocacy groups, in particular those aligned with various space-related industries, often have the capacity to shape and drive public policy, that influence is not one-dimensional. Groups, even powerful ones, are not always in control of their own destinies, nor can they assume that their interests will be reflected in public policy decisions. If this were so, public policy would look a bit different than it often does. For one thing, if advocacy groups had the clout so often presumed of them, the space program throughout the 1990s probably would have been larger and more central to United States (US) policy-makers' concerns. It is useful to note the significant growth over the past few years in the budgets of almost all federal research and development (R&D) agencies except for NASA.

The array of interests surrounding the space program should play dominant roles in shaping national decisions on space. This web of interests includes the nation's major military and space contractors, their affiliated labor unions, the suppliers connected to these industries, an entire scientific and technical community, and a wide array of groups focused on broad foreign and national security issues. Taken together, these groups certainly represent potent economic, societal, and political interests. So why is it that these groups have relatively little clout in the formulations of space policy?

Maybe that's not the correct question. Possibly the correct focus should be on the kinds of space policy issues that are open to group influence and those that are not. A rough analogy to defense policy is appropriate here. Defense industries may have a lot of influence in Congress and the Department of Defense (DOD) on decisions involving choices among competing weapons systems. But do these groups really shape "defense policy" any more than they shape "space policy?"

To understand this question, this chapter examines the notion of space policy "advocacy coalitions" and the role they play in defining the issues

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 61 Atwill, *Fire and Power*; Penley, *NASA/TREK*; and Dean, *Aliens in
 America*.
 McCurdy, *Space and the American Imagination*.

CHAPTER 2. RATIONALES OF THE SPACE PROGRAM

1 "Ancillary policy does not set out to solve an identified national
 problem. Policy in its most concrete form consists of a plan of action
 designed to solve an identifiable problem. But ancillary policy is
 more apt to represent a continuing government commitment. It does
 not necessarily solve a current problem, even though the bureaucracy
 it maintains may have been set up for that purpose long ago. Ancillary
 policy has low agenda status; it receives only limited public attention,
 public funds, and efforts of public officials. Media coverage, public
 opinion polls, legislative debates, and Presidential communication
 about the matter are sporadic at best. It is a matter of modest
 continuing importance that generates little interest among policy
 makers concerned about long-term directions or goals." Lyn
 Ragsdale, "Politics Not Science: The US Space Program in the
 Reagan and Bush Years," in Roger D. Launius and Howard E.
 McCurdy eds., *Spaceflight and the Myth of Presidential Leadership*
 (Urbana: University of Illinois Press, 1997), 135.

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 Program* (Urbana: University of Illinois Press, 1964), Part II.
 3 In 1997 the commercial space sector, for the first time, was greater
 than the governmental civil space sector as measured by expenditure
 of capital.

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- 12 McDougall, ...*The Heavens and the Earth*, 394-95.
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CHAPTER 3. ADVOCACY COALITIONS AND SPACE POLICY

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